

MORA: AI-Mediated Story-Based practice for Speech Sound Disorder from Clinic to Home

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Figure 1: *MORA* is a home-based practice tool for children with Speech Sound Disorder (SSD), formatted by an interactive story and conversation. *MORA* encourage the child to practice the target sound in complex forms, such as words and sentences in narratives (Center). Enabling home-based practice within natural contexts reinforces the generalization of speech sound (Right). Furthermore, *MORA* forms the AI-in-the-loop in the SSD therapy session. Speech-language-pathologist (SLP) configures the story with target sounds (Left), and the child practices at home with the aid of caregivers. SLP asynchronously monitors the progress and can adjust the therapy goal.

Abstract

Speech sound disorder (SSD) is among the most common communication challenges in preschool children. Home-based practice is essential for effective therapy and acquiring generalization of target sounds, yet sustaining engaging and consistent practice remains difficult. Existing story-based activities, despite their potential for sound generalization and educational benefits, are often underutilized due to limited interactivity. Moreover, many practice tools fail to sufficiently integrate speech-language pathologists (SLPs) into the process, resulting in weak alignment with clinical treatment plans. To address these limitations, we present *MORA*, an interactive story-based practice system. *MORA* introduces three key innovations. First, it embeds target sounds and vocabulary into dynamic, character-driven conversational narratives, requiring children to actively produce speech to progress the story, thereby

creating natural opportunities for exposure, repetition, and generalization. Second, it provides visual cues, explicit instruction, and feedback, allowing children to practice effectively either independently or with caregivers. Third, it supports an AI-in-the-loop workflow, enabling SLPs to configure target materials, review logged speech with phoneme-level scoring, and adapt therapy plans asynchronously—bridging the gap between clinical sessions and home practice while respecting professional expertise. A formative study with six licensed SLPs informed the system’s design rationale, and an expert review with seven SLPs demonstrated strong alignment with established articulation-based treatments, as well as potential to enhance children’s engagement and literacy. Furthermore, discussions highlight the design considerations for professional support and large configurability, adaptive and multimodal child

interaction, while highlighting MORA's broader applicability across speech and language disorders.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → Education.

Keywords

Do, Not, Us, This, Code, Put, the, Correct, Terms, for, Your, Paper

1 Introduction

Speech sound disorders (SSD) is one of the speech and language disorders characterized by having challenges in producing speech sounds accurately (e.g., saying “wed” instead of “red”). SSD affects roughly 5% of U.S. children and represents one of the most frequent communication challenges in preschool populations [19, 24, 64, 80]. These difficulties can affect speech intelligibility and, if left untreated, can persist into later childhood. In 8-year-old children, 3.6% were estimated to have persistent SSDs [99]. Speech-language pathologists (SLPs) play a central role in the prevention, assessment, and intervention for SSD [4]. Common intervention framework includes articulation- and phonological-based approaches [18, 97]. Prior classifications [98] highlight a range of methods: environmental, auditory-perceptual, cognitive-linguistic, production-based, and integrated interventions. A major therapeutic goal in SSD treatment is sound generalization, the transfer of stabilized sound production from structured practice to sound combination, facilitating carryover of sound productions at increasingly challenging levels such as syllables, words, phrases/sentences, and conversation speaking [10].

However, SLPs' availability is often limited, with typical sessions lasting only 30-60 minutes and happening only once or twice per week. Therefore, home-based practice plays a crucial role in reinforcing sound generalization beyond the clinical setting [83-85, 92, 92], increasing the opportunities for practicing sound productions. In addition, some children with SSD experience social communication difficulties [53, 100]. Since they tend to engage more frequently in conversation with parents or siblings [63, 85], home-practice can also serve as a meaningful opportunity to promote both generalization and social interaction.

Despite its importance, home-based articulation practice poses several challenges [82]. First, caregivers often lack professional training, making it difficult to provide accurate feedback or model correct productions [85, 95]. Inaccurate practice may reinforce incorrect speech patterns, hindering progress. Second, caregivers frequently report difficulty maintaining their child's motivation [85], or cite barriers such as limited time and fatigue [43]. Sustaining children's engagement in repetitive, routine-based activities can also be particularly challenging [83]. Recent digital tools aim to bridge this gap through gamified and story-based environments (e.g., Apraxia World [46], Spokelt [11], Little Bee Speech [7], and Apraxia Farm [1]). Such a system can increase practice opportunities and engagement, thereby promoting generalization [27, 62]. Story-based interventions provide meaningful contexts that encourage connected speech production [37] and can simultaneously support emergent literacy [59, 96], which often develops in parallel

with speech [36, 88]. However, most existing story-based interventions remain largely static and unidirectional, offering limited interactivity or feedback integration between home-based practice and SLP-guided therapy. This lack of dynamic engagement and professional feedback weakens their alignment with evidence-based treatment process.

To address these limitations, we present *MORA*, an interactive story-based practice system that enables children with SSD to practice target sounds through dynamic, character-driven conversations. By embedding sound targets within responsive dialogue, *MORA* encourages natural repetition and generalization across linguistic levels. Furthermore, *MORA* introduces an AI-in-the-loop workflow which allows SLPs to set up materials, monitor progress, and adjust therapy plans asynchronously, ensuring professional oversight without requiring synchronous supervision, respecting their expertise [52]. A formative study with six licensed SLPs aided in constituting the design rationales of *MORA*. Beyond promoting speech generalization, the conversational narrative format also fosters social communication [37, 96]. Through turn-taking and perspective-taking with story characters, children engage in pragmatic and expressive language practice in naturalistic, socially meaningful contexts—skills often related to, but distinct from, articulation ability.

We evaluated *MORA* with seven licensed SLPs in the United States to examine clinical validity and integration potential. Results indicate strong alignment with existing therapy practices, perceived benefits for monitoring and data-driven assessment, and broader potential to extend story-based conversational approaches toward supporting literacy and social communication development.

The contribution of this work is fourfold:

- Findings from a formative study and prototype walkthrough with six licensed SLPs, uncovering current challenges in story-based SSD intervention and home-practice monitoring, along with the current best practices.
- *MORA*, an interactive story-based practice system that supports speech sound generalization through character-driven narratives, caregiver-led home-practice, and a data-driven therapy system that supports monitoring and feedback.
- Insights from expert evaluation with seven SLPs, demonstrating *MORA*'s effectiveness for speech generalization at multiple linguistic levels, integrity, and perception on the AI-mediated system.
- Design consideration for the configurability and professional support in AI-in-the-loop system, adaptive and multimodal child interaction, and generalizability across speech and language disorders.

2 Background

2.1 Intervention and Generalization of Speech Sound Disorders

Children with SSD often face persistent difficulties in producing speech sounds, understanding language, and engaging in effective communication. Clinical interventions typically rely on structured methods such as imitation, drills, and minimal contrast therapy,

which involve practicing target phonemes or words through repetitive, controlled tasks guided by SLPs [79, 98]. These interventions are evidence-based and widely used to improve communication skills. However, the effectiveness of therapy depends not only on in-session performance with a clinician in the therapy room but also on the generalization of skills to real-world settings (e.g., home, community) with different speakers (e.g., peers, parents, teachers) [13]. Due to the speech sound complexity [27, 73], generalization refers to the child’s ability to apply practiced skills across different contexts and communicative situations, and demands high-frequency home-practice for consistency. Research consistently shows that intensive, distributed practice, including home-based exercises, enhances generalization and long-term outcomes. Parent-implemented and home-based interventions, especially when supported by digital tools, increase practice opportunities, intervention intensity, and child engagement, all of which are linked to better generalization [27, 62]. Studies highlight that adapting home-practice to the child’s context, providing individualized tasks, and involving caregivers in therapy planning are key for successful generalization [55–57]. Yet, families often struggle to implement such activities consistently due to a lack of professional expertise and daily life constraints [85]. Furthermore, the increasing caseloads and limited availability of SLPs exacerbate these challenges, limiting children’s opportunities for individualized attention [21, 30]. This highlights the need for tools that not only provide high-quality practice but also support continuity between clinical sessions and at-home environments. Bridging this gap is a core goal of many recent digital intervention efforts [49, 62, 79].

2.2 Scaffolded Speech Sound Intervention Through Hierarchical Practice

Speech sound development in children typically progresses through a developmental hierarchy—from phonemes to words, then to phrases, sentences, and full conversations [41, 73]. To target speech sounds at the phoneme level, the initial focus is on accurate production of target sounds in isolation, using techniques like auditory discrimination, imitation, and phonetic placement [27, 81]. Once accuracy is achieved in isolation at the phonemic level, practice moves to words starting with simple syllable structures and progressing to more complex ones; therapy activities may include minimal pairs, word repetition, and word games [25, 49, 98]. Beyond words, for the phrase/sentence level, SLPs then target the sound in short phrases and sentences, increasing linguistic complexity and promoting generalization [27, 41, 49]. Lastly, the final stage at the connected speech and narrative level involves practicing target sounds in spontaneous conversation, storytelling, and narrative tasks to ensure carryover to real-life communication [27, 41, 49].

Instructional strategies that reflect this progression have been shown to support more effective language learning and long-term generalization [13]. However, many digital applications and interventions focus narrowly on phoneme- or word-level exercises, lacking the scaffolding needed to transition into more naturalistic communication contexts [35, 40, 46, 47, 77]. Narrative-based learning has gained traction as a strategy to scaffold this hierarchy [38]. Stories create meaningful contexts that allow children to practice

speech targets in functional, engaging scenarios. For example, Little Bee Speech integrates storytelling with structured phonics and word practice [7], while WordWeb uses a language hierarchy as its backbone [14]. Working on narrative skills further can enhance children’s overall language literacy skills for long-term benefits [96]. Yet these tools often require extensive manual preparation by SLPs or caregivers, limiting their scalability and adaptability to each child’s needs.

2.3 Digital Tools Supporting Speech Therapy

Over the past decade, a wide array of digital tools has emerged to support speech-language therapy for children [16, 33, 39, 45, 48, 76, 93]. Some applications developed by clinicians [7, 33, 48] digitalize conventional paper-based therapy stimuli, while other applications like Apraxia World [46] and SpokeIt[11] further applied gamification techniques to encourage repetition and sustain engagement. Others, such as the intervention designed by Ansari et al. [14], incorporate multisensory input (e.g., touch, audio, visual cues), which aligns with best practices in speech and language learning. These tools are particularly effective at supporting phoneme and word-level practice through drill-based formats. Recent developments in Generative AI and automatic speech recognition (ASR) have further expanded the design space for speech therapy applications. AI agents can simulate conversation partners, adapt content in real-time, and reduce the manual burden on SLPs [32–34, 79, 86]. Additionally, ASR systems fine-tuned on disordered speech have shown improved accuracy in recognizing children’s speech [91], making it feasible to build responsive, child-facing tools that provide instant feedback [31].

Despite these advances, most tools remain limited in their ability to support generalization, particularly in transitioning children from isolated drills to functional communication. Moreover, few systems effectively integrate with SLP workflows, missing the opportunity to create therapeutic feedback loops between home and clinic. Addressing these limitations requires systems that are not only technically capable but also designed around the realities of clinical practice and child development.

3 Formative Study

To fill the blank of understanding current practices and challenges in SSD therapy and home-based practice, we conducted a formative study configured by two sections: semi-structured interviews and feedback on an early-stage prototype. The main information we want to gather from the interview includes (1) current practices and challenges in the general SSD clinical process, (2) how SLPs generalize the speech sound and design the corresponding contents, especially story-based, (3) the way SLPs handle home-practice and monitoring, and (4) how SLPs currently employ technology in therapy.

The prototype walkthrough aims to explore the design space of how speech practice can be embedded in the basic format of an interactive story, and identify concrete design rationale. The study protocol was reviewed and approved by the IRB board at our institution.

Alias	Gender	Age	Work Experience (years)	Employment Type	Work Setting
P1	Female	20–29	1–5	Mainstream Primary School	Mainstream Primary School
P2	Female	20–29	1–5	Pediatric Outpatient Clinic	Clinic
P3	Female	30–39	1–5	Private Practice	Clinic
P4	Female	50–59	11–15	Pre-K-22 public school	Pre-k- Turning 22 program
P5	Female	20–29	1–5	AI Services company/ Private practice	Clinic
P6	Female	30–39	1–5	Mainstream Primary School	Mainstream Primary School

Table 1: Formative study participants’ demographic data

3.1 Participants

We recruited the six participants through referral. All participants are based in the United States and have a Certificate of Clinical Competence in Speech-Language Pathology (CCC-SLP). The five participants have 1–5 years of working experience as a licensed SLP; one has 11–15 years of experience. They mostly worked in a clinic or a mainstream primary school. We provide more details about the participants in 1. The average level of developing home-practice is 2.6 on a 5-point Likert scale. The average of exploiting technology in the clinical content is 4.0 on a 5-point Likert scale. The primary tech-based services they mentioned to use are chat-interface LLMs, several apps for activities, and YouTube. The device mentioned is the iPad. The list of questions of the demographic survey was referenced from Ansari *et al.* [15]. We compensated each participant with a USD \$50 gift card for their participation.

3.1.1 Protocol. The study consists of a semi-structured interview and a prototype design walkthrough. At the beginning of the study, a researcher collected the participants’ informed consent forms and demographic information. Starting from the general clinical process and challenges of SSD, we deep-dived into the patterns from interventions, frequently used therapy methods/theory on each step of speech sound generalization, and the specific circumstances and challenges of home-practice. We obtained contextual information alongside a demonstration of the actual material being used. Following the 30-minute interview, we showcased an early proof-of-concept prototype to facilitate the home-practice. The interactive story format prototype targeted /ch/-initial sound, consisting of a single dialogue with the character *champion*. The scenario that the SLP assigns home-practice to a child, and the child does practice, is explained by the researcher as a verbal before presenting the prototype. We asked SLPs about the clinically desired direction of home-practice in this scenario, potential opportunities for AI integration, and any possible risks. The study lasted about an hour and was carried out remotely over Zoom.

3.2 Analysis

All interviews were audio-recorded and later transcribed. Applying established thematic analysis methods [22, 23], one researcher open-coded the transcript to identify emerging themes. The whole research team then finalized the themes through discussions. The following section 3.3 represents the identified findings.

3.3 Findings

3.3.1 Finding 1: *Story-based intervention for SSD therapy has been employed with advantages, but the materials lack interactivity.* Story-based interventions were typically introduced after children demonstrated independence with target sounds and were ready to begin generalization. Because these materials incorporate short sentences and contextualized speech, they were described as more suitable for older children—including those in mid-primary school—who are working on later developing or residual sounds. As P1 explained, “Whereas a lot of school-aged kids, like maybe 8, 9 years old, they would mostly come in working on more of the later sounds if they haven’t developed them yet, or you know, earlier sounds if they still haven’t mastered them yet.” Clinicians emphasized that story materials are intentionally structured around repeated target sounds, providing opportunities for varied practice within meaningful linguistic contexts. P2 noted, “They (the provider of story-intervention materials) have like stories where the S is repeated multiple times in that story.” Such repetition allows children to reinforce target articulations while practicing connected speech at the phrase or sentence level. Participants also highlighted that story-based practice can facilitate educational and communicative development once children move beyond isolated words. As P4 described, “A lot of times, we use them after they can produce it at the word level. So when we move to the sentence level or the conversational level, we always want to integrate literacy, especially in the educational part, because in order to qualify for speech services in schools, there needs to be an educational impact. So that means either academic or social.” This reflects how story-based materials naturally bridge therapeutic goals with literacy and communication contexts, aligning with the educational priorities of especially school-based SLPs.

Despite these advantages, a major challenge identified in current story-based interventions is getting children’s engagement, particularly given their limited attention spans. As P2 explained, “[A story website they use], it’s just like a paragraph so that I can read it, but the kids are like it’s just like lots of talking, so they can get disengaged.” Nonetheless, they tried to make sessions more dynamic, but the lack of built-in interactivity remained a limitation. P2 continued, “So I try to draw it sometimes or make it a little more engaging by acting it out. It can be a little disengaging because it’s like a slightly long story with no pictures or anything, just words.” To compensate for these limitations, some clinicians incorporated external multimedia sources such as animated stories to capture children’s attention. As P5 shared, “[I sometimes] use animated story on Youtube”. These accounts demonstrate how the absence of interactive components

in existing story materials hinders sustained engagement, underscoring the need for more multimodal and participatory formats in home-based practice.

3.3.2 Finding 2: *Keeping children engaged in therapy is the main challenge of repetition.* Since a significant part of SSD therapy involves repetition and *drill-based* [69], maintaining children’s attention during repeated productions of the same sounds was identified as a persistent challenge. P2 summarized this difficulty: “*I think the most common challenge is that the kids lose interest, because it’s such drill work, nothing other than practice to get that sound in.*” To sustain engagement, clinicians frequently incorporated interactive or game-based elements into traditional methods, such as flashcards. P1 described using tactile materials: “*I usually try to use something more interactive. I have one where they can physically remove Velcro cards and move them, because they’re more engaged when it’s interactive.*” P3 explained a gamified flashcard, “*My go-to is always like the flashcards. For example, I’ll play a game (...), and each turn they take, they pick up a flashcard, and we do three to five articulating trials. After finishing, the child gets a turn in the game.*” Similarly, P2 noted integrating the repetition into familiar board games: “*We’re playing something like Chutes and Ladders, Ticket to Ride, or any others. Introduce the sounds, they roll the dice and get five, they’ll say the sound five times.*” While these approaches effectively captured children’s attention, participants acknowledged a trade-off between engagement and practice intensity. As P2 reflected, “*(Introducing how to gamify the therapy process ...) When I do that, it reduces the number of times I can get the word in practice.*” Given the limited duration of most therapy sessions, this reduction in production frequency can significantly affect overall treatment dosage, which we should also consider in home-based practice design.

3.3.3 Finding 3: *Data-driven support needed for granular dynamic assessment in speech therapy.* Participants emphasized that speech therapy relies on continuous, fine-grained assessment, but current systems lack tools to capture such detailed client progress. At different stages of generalization, clinicians monitor how accurately a child produces target sounds and the amount of cueing required. P6 illustrated this process: “*This time, the kid can produce /s/ and /z/ sounds at the freeze level with no more than two queues, about eight out of ten times. The next level is sentence production, so reduce the cueing.*” Such iterative monitoring and goal adjustment were described as the core of clinical decision-making. As P6 explained, “*Clinical treatment is a dynamic assessment of progress. You don’t do a standardized assessment all the time, but you do an informative assessment all the time.*” Beyond the clinical tracking, SLPs affiliated with schools or institutions must also report periodic progress. P1 mentioned, “*For different school districts, they have different requirements. (...). But all school districts ask you to do a progress report after a certain period.*” To meet these reporting and diagnosis needs, clinicians desire systems that automatically collect and summarize client data over time. Participants also highlighted the specific value of capturing speech samples from home-practice, as children’s performance often differs outside therapy sessions. P5 remarked, “*They might be doing really well with me in the treatment room, but it doesn’t sound the same when they’re outside and practicing at home.*” Access to home audio data would allow clinicians to evaluate generalization more accurately and adapt intervention

plans accordingly, reinforcing the need for data-driven support in dynamic assessment.

3.4 Design Rationales

We address the key design rationales derived from the prototype walkthrough and the findings of current practices. These three design rationales guide the final design of *MORA*.

3.4.1 DR1: *Interactive Story Practice Using Role-Play in the Narrative.* Findings from Section 3.3.1 indicated that while story-based interventions effectively engage children, their interactivity often reduces sustained attention. To address this gap, we drew inspiration from an SLP’s use of role-playing as a therapeutic technique to increase connected speech. As P5 shared, “*I would say role-playing a lot, especially with the older kids in general, even with language, speech, and stuttering, because that is more unstructured, (...) If they’re just talking in conversation, they can easily mispronounce.*” Role-play provides a naturally conversational and less constrained setting, encouraging children to produce speech sounds in authentic communicative exchanges. Building on this insight, we design an interactive story practice where children assume roles within a dynamic narrative. Instead of passively listening, they dialogue with story characters, responding verbally to move the plot forward. Prompting each response to include target sounds provides repeated opportunities for production within meaningful contexts that mirror conversational use. We also incorporated a *madlib*-inspired activity to support Word Mode, a bridge between isolated word practice and connected sentence production. P1 explained the Madlibs: “*Madlibs basically are given a story with a bunch of blanks and you can fill in the blanks with words starting with a target sound. The Madlibs activity was quite engaging and has a possibility for a story format.*” In this mode, children fill in missing words within the story sentences, embedding target sounds into a varied linguistic environment while maintaining narrative relevance. This structure allows for systematic generalization from word-level to sentence-level practice. Overall, this design rationale merges narrative immersion, structured sound repetition, and interactivity, extending traditional story-based therapy into an active, conversational learning experience that fosters both engagement and speech sound generalization.

3.4.2 DR2: *Support cues, instructions, and feedback in multiple modalities.* Participants emphasized the importance of multisensory cueing in sustaining children’s attention and supporting accurate speech learning, consistent with prior work on technology-assisted therapy [14]. Parents are typically responsible for providing auditory or visual cues in home-based practice, yet their involvement and cueing consistency vary. Since most caregivers are not trained specialists, built-in visual and auditory supports can empower practice and maintain effectiveness [28]. Visual prompts were essential for demonstrating articulator placement—such as tongue position or lip rounding—and supporting children’s imitation [29, 74]. As P1 described: “*I always have the mouth model with me because I use it so much. It’s usually for dentists, but I bought it and I removed the tongue. So that way you can move the tongue to show tongue placement for certain sounds.*” In addition to mouth models, SLPs use phonemic hand cues [75], linking gestures to specific sounds.

P5 illustrated: “I’ll have a tactile component like for *S* I love doing it on my arm and I have the kid trace on the arms.” These tactile–visual gestures create an embodied connection between movement and phoneme production, reinforcing learning through motor feedback. Audio was also considered a critical modality for instruction and self-correction, particularly for pre-literate children. Voice-based guidance allows children to follow tasks without reading, while listening back to recordings supports self-monitoring. P1 suggested, “Listen back and hear what they said also would help even if you don’t have that accuracy portion.” SLPs also emphasized repeated auditory exposure—auditory bombardment—as an effective priming technique. As P6 explained, “(Keep exposed to the target sound in the story) we call that auditory bombardment. So people will hear our sounds pretty often. It’s pretty helpful, a good prime they can.” This repeated exposure sensitizes children to target sounds before active production. Overall, these insights motivated us to design a system that integrates visual and auditory cues into story-based interventions, ensuring children receive multimodal guidance and feedback that aligns with clinical practices.

3.4.3 DR3: Supporting granular progress monitoring through layered visualization and clinician control. Findings in Section 3.3.3 underscored clinicians’ ongoing need to collect and interpret detailed client data for dynamic progress monitoring. Based on this, we established the design rationale that **MORA** should enable SLPs to efficiently interpret and monitor children’s progress through in-app recordings and structured summaries. Participants expressed the need for two complementary levels of support: statistical overviews to capture general progress patterns and transcriptions to conduct detailed diagnostic reviews when necessary. Because therapy time is limited, listening to every recording is impractical; however, clinicians still desire on-demand access to specific cases to inform goal planning and reporting. Participants emphasized the importance of maintaining expert supervision, coupled with the current speech sound assessment model’s technical immaturity [60]. They viewed automated scoring as a supportive reference, not a replacement for clinical judgment. As P4 explained, transcripts reveal contextualized errors that simple scores cannot capture: “Both scores and transcripts are useful, but transcripts are particularly valuable because they reveal where errors occur in context. For example, if a child says ‘walk it’ without producing the final /k/, I can identify difficulty with word-final consonants. Similarly, if I notice an error in a phrase such as ‘S into rainbow,’ I can design practice phrases that end in /s/ and begin with /r/. Over time, I may also observe the need to target more complex sounds, such as the vocalic /r/.” Such contextual interpretation allows SLPs to refine therapy targets and adapt treatment sequences more precisely. These findings underscore the need for layered visualization and clinician-controlled navigation, enabling SLPs to transition seamlessly between summary dashboards and detailed transcription views. This layered access supports efficient monitoring, reinforces professional autonomy, and ensures that automated analysis enhances rather than replaces expert decision-making.

4 MORA

Building on the formative insights and design rationales identified in Section 3, we developed **MORA**, an interactive, AI-mediated

story-based system designed to extend speech sound therapy from clinic to home. The system was inspired by SLPs’ clinical strategies, such as role-play, multisensory cueing, and continuous progress monitoring, into a cohesive digital experience for at-home practices. **MORA** integrates interactive storytelling with real-time speech feedback, multimodal visual and auditory cues, and a clinician-facing dashboard for asynchronous monitoring and adjustment. Together, these features operationalize the principles of interactive, data-informed, and expert-in-the-loop therapy to support speech sound generalization and holistic child development.

4.1 Key Features

4.1.1 KF1: Overall Interactive Story structure. The interactive story practice embeds target sounds directly into the narrative by assigning them to characters’ names, locations, and items. This design ensures repeated exposure while maintaining narrative coherence. Children are instructed to produce utterances containing target words during conversational exchanges with characters. If the required words are omitted, the system provides immediate corrective feedback through voice prompts, encouraging children to attempt the production again (see (b:5) in Figure 2). Since the expected age range of this system would be early primary school age, each story session is designed to last approximately 8 minutes, a duration chosen to balance attention span with opportunities for repetition. Beyond following a linear storyline, children influence the narrative by making choices—such as selecting a destination or interacting with objects—which introduces a tactile dimension to the activity and affords them agency in shaping the story (a:7 in Figure 3). To support varying levels of practice, the system provides two complementary modes. Children complete partially structured sentences in the *word-mode* by producing target words in isolation. If they complete filling in, the children can listen back to the whole sentence. In the *sentence-mode*, they are guided to produce complete sentences incorporating the target words, allowing for more flexibility, making it more similar to an actual conversation. These features operationalize the design rationale by embedding repeated sound practice into engaging, interactive story contexts that combine visual, auditory, and tactile elements.

4.1.2 KF2: Features Motivated by Current Clinical Practices. The system operationalizes these clinical insights by incorporating established therapy techniques into the interactive story design, including the multisensory features discussed in Section 3.4.2. First, drill practice is embedded into the narrative structure, motivated by findings in Section 3.3.2. This allows children to encounter and repeat target words within the story. For additional practice, a dedicated target word sidebar provides opportunities for focused word-level production (a:2 in Figure 2). Second, auditory bombardment [26], one of the sound perception training treatments, is implemented by repeating target sounds in character dialogue. Target words are highlighted in the dialogue and linked to the sidebar, where children can replay pronunciations on demand (a:2-1 in Figure 2). Third, visual cues are supported by pictorial representations of mouth shapes for each target sound, accompanied by simple descriptions of articulator placement (a:3 in Figure 2). These visual prompts reinforce accurate production and guide parents in

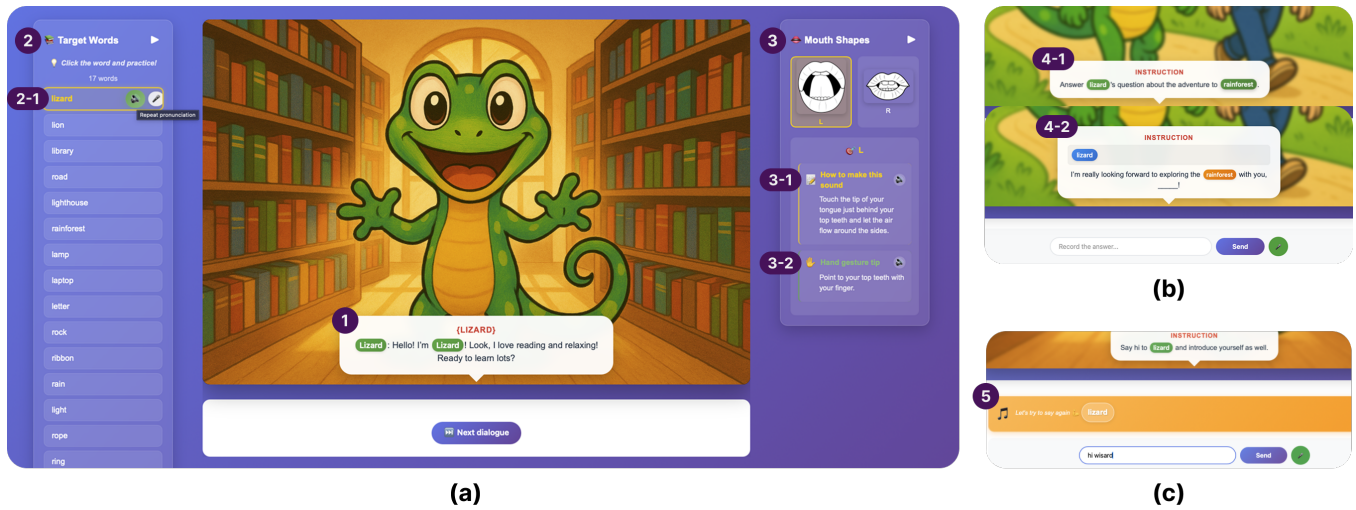


Figure 2: Main screen of MORA. (a:1) is the character's dialogue, which asks questions, has together, describes situations, and suggests interaction. Target words are highlighted as clickable green buttons that activate the information in both sidebars. (a:2) Target Words sidebar contains all the words in the story, highlighting the words in the current dialogue. (a:2-1) Each word card has repeated listening and pronunciation practice. (a:3) Mouth shape sidebar consists of a corresponding pictured mouth, how to make the sound, and a hand gesture tip for each sound, with voice support. (a:3-1) The way of making the mouth shape and placing the tongue is supported, and (a:3-2) how to make the phoneme hand gesture is described. (b:4-1) Instruction for children in *sentence-mode*. (b:4-2) Madlib activity at a sentence of children's side in *word-mode*. If the child finishes the sentence, the sentence is spoken. (b:5) If the child misses the word, feedback is delivered.

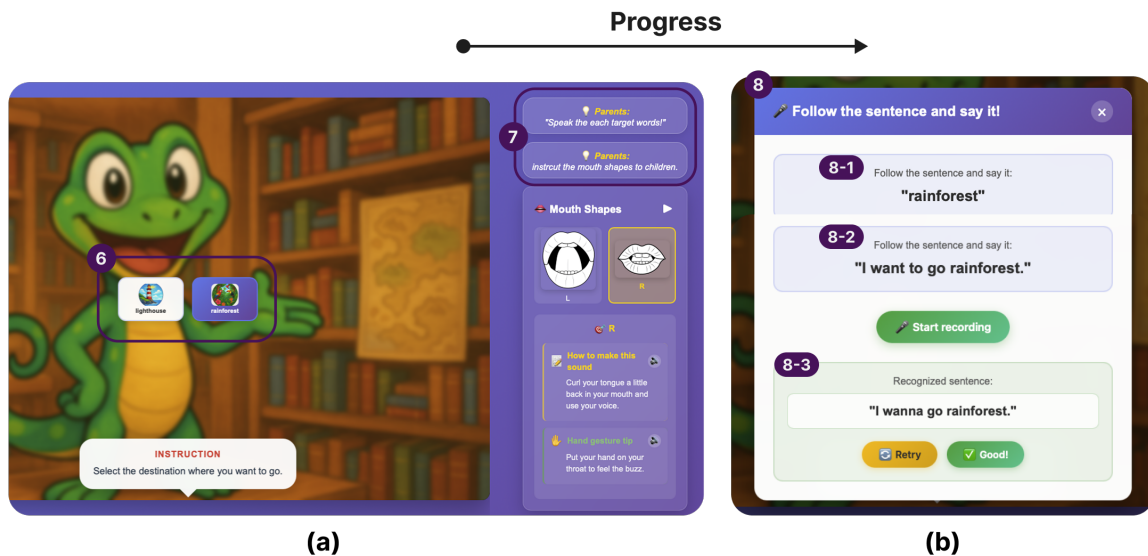


Figure 3: Additional features in MORA. (a:6) Children can choose the destination, which affects the narrative. Additional clickable interactions are inherited in the story. (a:7) Short instruction where parents can support the practice. (b:8) The prompt boxes lead production in the interaction, and supplemental practice in the Target Word sidebar. (b:8-1) The word is suggested in *word-mode*, and (b:8-2) the sentence in *sentence-mode*.

supporting practice at home. And the description of the gesture corresponding to each sound can be matched with phonemic hand cues commonly used in therapy. These approaches can stimulate sound

production. Finally, the system adapts the idea of madlib-style fill-in-the-blank tasks for word-level practice (b:4-2 in Figure 2). Children insert target words into prepared sentences by pronouncing them, and upon completion, the system reads the sentence aloud, bridging

isolated word practice toward sentence-level generalization. These features bring established therapeutic methods into an engaging, multimodal story environment grounded in clinical practice.

4.1.3 KF3: Feedback and Parental Involvement. The system also integrates mechanisms to provide feedback to children. When a child omits or mispronounces the target word, the system first prompts them with a voice instruction to try again. If difficulties persist, the child can attempt up to two retries, during which they are shown the transcription of their recording as an additional cue (b:5 in Figure 2, b:8-3 in Figure 3). Children can also listen to their own recording, supporting self-monitoring by recognizing the differences between their production and the intended target. At the same time, the system involves parents in the practice process. It helps caregivers reflect the clinical insight that children make the most significant progress, but also prevent the risk of frustration caused by AI oversight [28]. To this end, a tip message box (a:7 in Figure 3) is displayed to parents at each dialogue turn, notifying them of opportunities to provide cues or reinforcement. These lightweight prompts help parents participate more effectively, even without specialized training. Together, these features balance system-provided feedback with parental involvement, creating an environment where children benefit from external guidance while developing autonomy in correcting their speech.

4.1.4 KF4: Data-driven and clinician-guided monitoring through Dashboard. Sections ?? and 3.4.3 emphasized clinicians’ need for continuous, data-driven assessment. **MORA** addresses this by providing a dashboard that aggregates and visualizes children’s recorded speech from story sessions. Each session summary includes automatically generated scores based on phoneme-level transcription, calculated as a distance measure from ground-truth phonemes and categorized (see Section 4.2.2) into four levels—Excellent, Good, Fair, and Need Practice. These scores offer accessible indicators of progress while acknowledging the limitations of automated diagnosis. The dashboard visualizes aggregate statistics such as the number of productions per target word and pronunciation quality across attempts for efficient review. When more detailed analysis is needed, SLPs can drill down to individual recordings, accessing the computed distance scores, system-generated and phoneme-level transcriptions, original prompts, and corresponding audio files. This layered structure supports both high-level overviews and context-sensitive analysis, allowing SLPs to efficiently plan and refine interventions while maintaining expert control over interpretation.

4.2 Implementation

4.2.1 Implementation detail. **MORA** is implemented as a web application with a Python Flask backend server. The dialogue content and scene images are generated using OpenAI’s GPT-4.1-mini, while real-time voice feedback is provided through OpenAI’s GPT-4o-mini-tts. OpenAI’s GPT-4o-mini-transcribe handles audio transcription, and phoneme-level transcripts are produced using Google’s Gemini-2.0 Pro. To support phonetic accuracy, we additionally employ a Python library, *eng-to-ipa*, that leverages the Carnegie Mellon University Pronouncing Dictionary [3] to convert English text

Word	HuTr	Model	Prediction	Hamming	Label
rate /ret/	[ɹɛrts]	Gemini-2.0	[fɛrps]	0.208	Good
		Ginic-gender-split	[fɛrt]	1.083	Fair
		XLSR-TIMIT-B0	[fɛrt]	1.083	Fair
biscuit /bɪskɪt/	[bɪs:ɪt]	Gemini-2.0	[pɜfɛkt]	1.292	Fair
		Ginic-gender-split	[bɛsɛt]	0.083	Excellent
		XLSR-TIMIT-B0	[ɹædɛt]	1.5	Fair
fish /fɪʃ/	[fɪs]	Gemini-2.0	[fɪʃ]	0.083	Excellent
		Ginic-gender-split	[sɒʃ]	0.792	Excellent
		XLSR-TIMIT-B0	[sɛks]	1.167	Fair
ojo /oˈdʒoː/	[o:tʰo:]	Gemini-2.0	[ʊtʰo]	0.125	Excellent
		Ginic-gender-split	[oʊtʰʊ]	2.0	Fair
		XLSR-TIMIT-B0	[oʊtʰɪ]	2.083	Need Practice

Table 2: Examples of PFER by each model, Hamming distance, and corresponding Label in the dashboard of MORA. The dataset contains disordered speech recordings of children. Word has orthography and ground truth phonemic transcription. Human Transcription (Hu) is a transcription from disordered speech containing errors, and transcribed by human annotators. Prediction is the value predicted by Model.

into the International Phonetic Alphabet (IPA). To ensure consistency in visual presentation, we create character sheets that include variations in angles and emotional expressions, which are referenced during the scene image generation process. Items in the interaction scenes (see Figure 6) are generated separately, allowing them to be reused as interactive buttons or embedded in relevant scene images.

4.2.2 Phonemic scoring process. To support sound production quality in the glance view, **MORA** uses a scoring process based on phoneme transcription. We first obtain a phoneme transcription of the target words. The transcriptions are then pre-processed to extract the basic phonemes by removing notations related to stress, length, tone, syllable breaks, and boundaries. Both the performance limitations of current phoneme transcription models and the developmental level of child speech justify this cleaning step. As an additional step, we normalize r-colored vowels, which are influenced by the rhotic “r” sound in American English [8, 9]. For example, the sequence ər can be replaced with the corresponding r-colored vowel ɜ.

For scoring, rather than treating phonemes as either completely identical or entirely different, we brought the metric PFER (phone feature error rate), whose basic unit is phone, but considering the phonetic similarities among phones with 24 phonetic features for each phone [65]. We employed the Hamming feature edit distance. Hamming distance calculates one feature mismatch cost as $1/24$, implemented in PanPhon [65]. This method assigns similarity-based costs to phoneme substitutions, allowing for more nuanced evaluation of the child’s pronunciation.

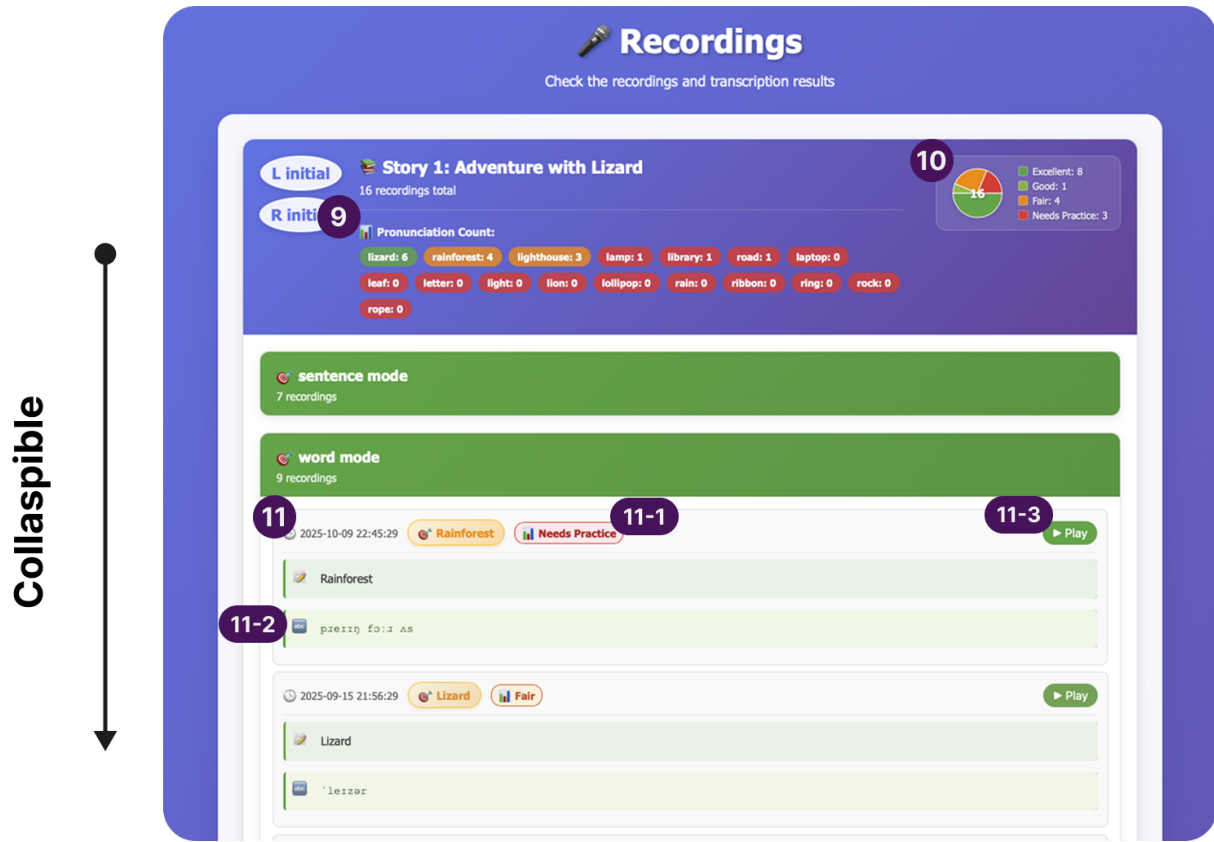


Figure 4: The screen of dashboard in MORA. (9) The number of pronunciations per target word. (10) Circle graph of aggregated production quality. (11) Each card contains the following information: (11-1) production quality label; (11-2) phonemic transcription for diagnosis by SLPs; (11-3) replayable audio clip.

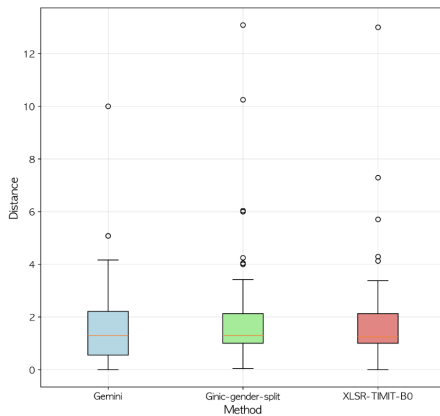


Figure 5: Boxplot of three phonemic transcription models, Gemini 2.0-Pro, XLSR-TIMIT-B0, and Ginic-gender-split on human annotated phonetic transcription (HuTr) and model predictions.

We compared three off-the-shelf audio-to-phoneme models, Gemini 2.0-Pro, XLSR-TIMIT-B0 [6], and Ginic-gender-split [5], to evaluate how they work in children’s voices, especially disordered children, and choose the proper model. We use the Disordered Speech

Errors database in the STAR database [12], configured by 162 audio clips recorded for each word, corresponding phonemic targets, and phonetic transcriptions. The boxplot in Figure 5 represents the trend of each model. Among the models, Gemini-Pro 2.0 ($M = 1.536$, $SD = 1.316$) demonstrated the most stable performance, with a consistent central tendency and lower variability than other models. Although XLSR-TIMIT-B0 ($M = 1.617$, $SD = 1.377$) and Ginic-gender-split ($M = 1.624$, $SD = 1.592$) achieved similar mean distances, their higher variance and extreme maximum values (up to 13 outliers in Hamming) indicate lower robustness to atypical or disordered productions. Table 2 represents an example of sound production quality labeling, the word with corresponding phoneme, Human Transcription (HuTr), and each model and their performance on each distance. These results suggest that Gemini provides the most reliable alignment between human and model transcriptions, and is stable enough to employ in disordered children’s audio clips. Considering that the comparison is conducted on phonetic and phonemic transcription, which needs additional converting, we defined four distance-based levels to categorize pronunciation quality: (1) Excellent (0.0–0.1), (2) Good (0.1–1.0), (3) Fair (1.0–2.0), and (4) Poor (>2.0).

5 Domain Expert Study

As SLPs can be considered to have two roles: experts in speech-language therapy and users who can employ *MORA* in their clinical practices, we conducted a domain expert study with 7 SLPs to evaluate *MORA*. The protocol of this study has been approved by the IRB board at our institution. The three research questions are as follows:

- RQ1. How do SLPs perceive the usability and integration of *MORA* within their existing therapy workflow?
- RQ2. How does *MORA* align with current SSD clinical practice?
- RQ3. How do SLPs perceive and experience the AI-mediated features of *MORA*?

5.1 Participants

We recruited participants using purposeful sampling through word-of-mouth (4 participants) [68] snowball sampling (3 participants) [44]. All participants identified themselves as female and are based in the United States. All participants hold a Certificate of Clinical Competence in Speech-Language Pathology (CCC-SLP); six participants had completed a Ph.D. degree, and one had an M.A. degree in the associated program. Four participants have 21+ years of working experience as licensed SLPs, one has 11–15 years, and two have 1–5 years. The participants' demographic information details are in Table 3. The average level of technology use in clinical content/intervention is 4.0 on a 7-point Likert scale. They mentioned multiple supplemental therapy materials, such as iPad apps, virtual stories, and videos. Participants were compensated \$60 USD for their participation. The questions of the demographic survey are adopted from Ansari *et al.* [15].

5.2 Study Design

Each study session lasted around 60 minutes and was conducted virtually over Zoom. Two-story materials, story template, and corresponding images were prepared. We employed a within-subjects experimental design to mitigate the bias. One is configured by target words with l and r initial sounds, and another is configured by s and r initial sounds. Two target sounds are arbitrarily chosen to show the variety in the study, and the words are automatically recommended by *MORA*. The details of the story in the study setting are described in Figure 6.

5.2.1 Protocol. At the beginning of each study session, participants provided informed consent and completed a brief demographic questionnaire. The researcher then introduced the study scenario, explaining that participants would evaluate *MORA* as if they were assigning it as a home-practice tool for a child working on word- or sentence-level generalization of target sounds. Participants were guided through the key features of *MORA* in both word and sentence modes before completing two questionnaires: a nine-item survey based on the Technology Acceptance Model (TAM) [94] and a fourteen-item survey assessing clinical relevance, intensity, balance, and educational impact. Both instruments used a 7-point Likert scale. Following the questionnaires, participants took part in a semi-structured interview designed to elicit deeper reflections on their experience and survey responses. Throughout the session,

the system automatically recorded screen activity and interaction logs for subsequent analysis.

5.2.2 Metrics and Analysis. We analyzed the collected data using a mixed-methods approach that combined quantitative and qualitative analyses. For the quantitative analysis, we computed basic descriptive statistics for each questionnaire item. Given the small sample size ($n = 7$), our analysis focused on identifying general response patterns rather than testing for statistical significance. To examine the consistency of the survey instruments, we calculated internal reliability using Cronbach's α [20], acknowledging that the results serve as reference values due to the limited number of participants. We further assessed expert agreement by computing the interquartile range (IQR) for each item, interpreting values ≤ 1 as strong consensus, 1–2 as moderate consensus, and ≥ 2 as low consensus [78]. Content validity was evaluated through the Content Validity Index (CVI) [54], defined as the proportion of participants rating each item above the midpoint of the scale, with values above 0.78 indicating high content validity [71, 72]. For the qualitative analysis, we employed open coding and thematic analysis [22, 23] on the transcribed interviews to iteratively uncover recurring themes, using ATLAS.ti [2]. Finally, we integrated the quantitative and qualitative results through triangulation analysis [90], highlighting areas of convergence and divergence between the two data sources.

5.3 Result

5.3.1 RQ1. How do SLPs perceive the usability and integration of *MORA* within their existing therapy workflow? Overall, SLPs evaluated *MORA* as both usable and readily integrable into their existing therapy routines. Post-questionnaire results revealed strong agreement that *MORA* would be a valuable tool in clinical practice and could be easily incorporated into workflows (see Figure 7). In the clinical practice survey (Table 4), participants rated the overall integration highly ($M=6.714$, $SD=0.787$), with strong consensus and a high CVI, suggesting consistent endorsement of the system's practical fit.

Bridging clinic and home through visualized client data. Participants highlighted the dashboard's role in connecting home-based practice with clinical oversight. They valued the automatically generated statistics, such as practice frequency and pronunciation scoring, as a low-effort way to monitor progress and set next goals, as they provided the essential data. As P4 noted, "*Those headers [scored labels] are good... helpful. That's easy to look at, grasp information from, and take away*". P1 appreciated the low effort of the dashboard, "*it's great to gather the recordings and have the transcriptions and the estimations... knowing that I can click them out and look at them, that's great*". This feature was reflected in the survey results ($M = 6.43$, $SD = 0.79$; see Table 4) and indicates that the dashboard improved the ability to monitor children's progress as well ($M=6.57$, $SD=0.79$; see Figure 7). Some SLPs suggested that further configurability could better align with their personal workflow preferences—a point discussed further in Section 6. Several participants also emphasized how the summarized statistics supported communication with parents: "*The app would make it easier to provide feedback to parents and children... based on very solid statistical analysis*" (P5). The ease of providing feedback to parents/children

Alias	Gender	Age	Degree	Work Exp.	Employment Type	Work Setting	# of SSD clients
E1	Female	50–59	Ph.D	21+ years	Self-employment	Client's home	30+
E2	Female	50–59	Ph.D	21+ years	Pre-K/Kindergarten	Mainstream Primary School	30+
E3	Female	60+	Ph.D	21+ years	Self-employment	Their Home/Office	30+
E4	Female	40–49	Ph.D	11–15 years	University/College	University/College	6 – 10
E5	Female	20–29	M.S.	1–5 years	Hospital	Hospital	21–30
E6	Female	50–59	Ph.D	21+ years	University/College	Literacy Based Preschool	30+
E7	Female	20–29	Ph.D	1–5 years	Self-employment	Private Practice	1–5

Table 3: Expert study participant demographic data

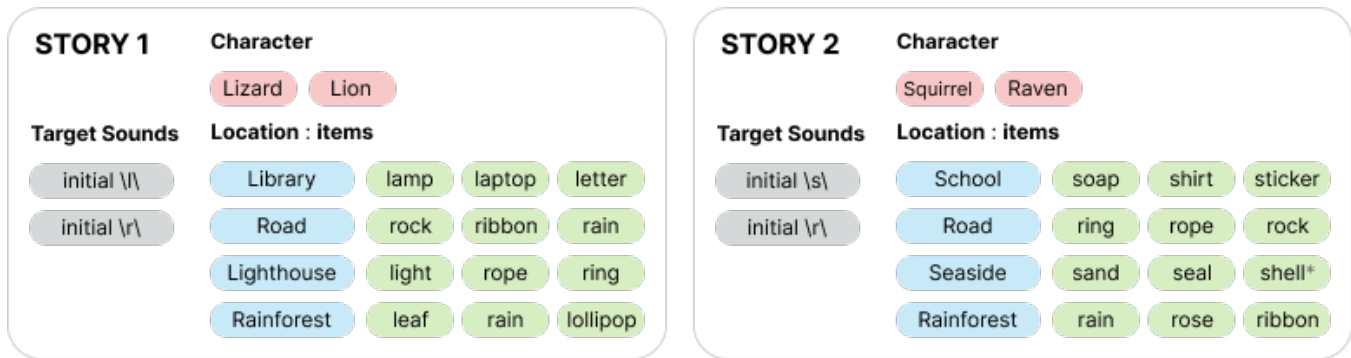


Figure 6: Configuration of Stories in Expert Study

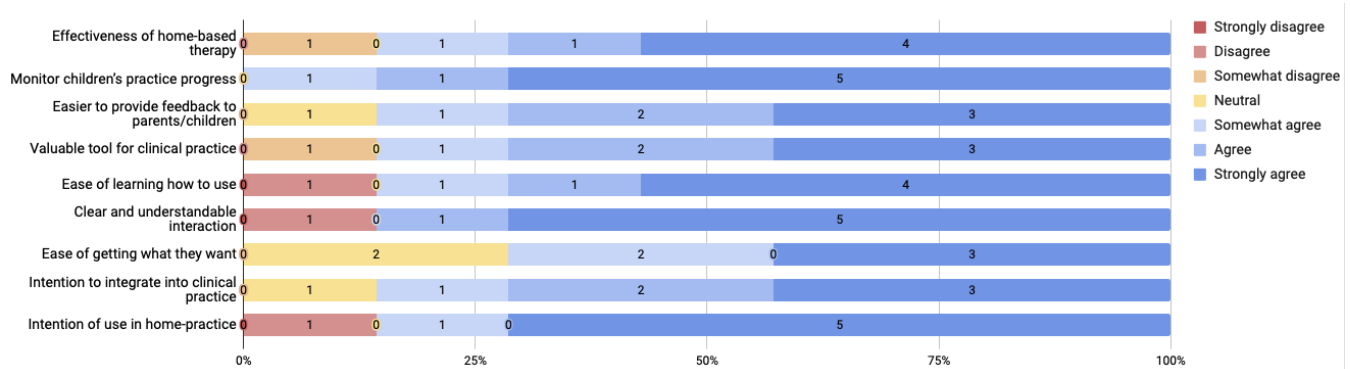


Figure 7: Post survey result: Technology Acceptance Model

is demonstrated through the TAM survey with an average of 6.0 (SD=1.15). These findings suggest that the AI-in-the-loop dashboard promotes transparency, professional trust, and continuity of care across clinic and home settings.

Flexible integration across home, clinic, and group settings. SLPs also saw potential to apply *MORA* beyond home practice—for instance, during therapy sessions or in group-based settings. They noted that *MORA* could support independent practice in therapy while automatically recording children's speech for later review. P3 illustrated that, "It would allow the additional practice, they'd be engaged and interested in doing it, plus you're getting a recording, which makes it a huge difference, too, being able to hear what the student actually said." Group therapy, commonly implemented in

school settings, can be a prime example of additional practice. Another participant added, "You can have a large group and have the students practicing independently. And then you go around and check them. Something like this would be perfect for that.", and continued, "At the end of the therapy session, I should say. Having a student just do some final practice. On their own, while taking, you know, data notes or, you know, session notes" (P2).

5.3.2 RQ2. How does *MORA* align with current SSD clinical practice? Overall, participants agreed that *MORA* closely reflects current evidence-based practices for treating SSD ($M = 6.0$, $SD = 1.20$; see Table 4). Their feedback highlighted how its design aligns with

Criteria	Mean	STD	Median	IQR	Consensus	CVI
Evidence-based Practices	6.000	1.291	7	2.0	Low consensus	1.000
Integrity into Therapy	6.571	0.787	7	0.5	Strong consensus	1.000
Sufficient Practice Intensity	6.714	0.488	7	0.5	Strong consensus	1.000
Balance of Practice & Play	5.857	1.069	6	1.0	Strong consensus	1.000
Drill/practice	6.000	1.155	6	1.5	Moderate consensus	1.000
Auditory Bombardment	6.286	0.756	6	1.0	Strong consensus	1.000
Madlib-style activity	6.571	0.535	7	1.0	Strong consensus	1.000
Visual Cue & Description	5.286	1.496	5	2.0	Low consensus	0.857
Minimal Pair, Multiple oppositions, Stimulability	4.000	1.414	5	2.0	Low consensus	0.714
Statistics Dashboard	6.429	0.787	7	1.0	Strong consensus	1.000
Audio Clips & Transcript	6.714	0.488	7	0.5	Strong consensus	1.000
Parental Involvement	6.143	1.464	7	1.0	Strong consensus	0.857
Literacy support	5.857	1.676	7	2.0	Low consensus	0.857
Social Communication support	5.143	1.215	5	2.0	Low consensus	1.000

Table 4: The result of the clinical survey, analyzed by descriptive statistics, interquartile range (IQR) with corresponding agreement, and content validity index (CVI). The Cronbach's α is 0.866.

key therapeutic principles—such as generalization, practice intensity, multimodal cueing, and contextualized learning—while maintaining engagement and clinical flexibility.

Narrative-based practice for generalization. Participants emphasized that the story- or conversation-based format effectively supports speech generalization beyond isolated word practice. They agreed that the effectiveness and motivational part of the interactive story format ($M=6.57$, $SD=0.54$) had strong consensus and a high CVI. They described how embedding targets within connected speech contexts helps children transfer learned sounds more efficiently: “In connected speech, we have assimilation where sounds influence other sounds, and we don’t always make our speech sounds the same naturally”. Following that, effectiveness contrasts with isolated repetition. “If we are just targeting single words, it takes a really long time to transfer and maintain those skills” (P6). This finding aligns with *MORA*’s design goal of facilitating generalization through interactive narrative contexts—an approach that mirrors naturalistic speech use in therapy and daily communication.

Balancing structure and engagement. Survey results indicated that *MORA* provides sufficient practice intensity—both in number of repetitions and engagement ($M = 6.71$, $SD = 1.07$)—while maintaining a balanced mix of structured practice and play ($M = 5.86$, $SD = 1.07$). SLPs valued how the conversational story format sustained children’s motivation while preserving therapeutic rigor: “Kids like stories because it doesn’t feel like they’re doing any work. I really like having them create their own sentences instead of just copying one” (P5). This balance demonstrates how interaction design can integrate structured articulation within intrinsically motivating play contexts.

Designing dynamic, multimodal cues. Participants emphasized the need for more dynamic and multimodal visualizations to better support speech sound learning. While visual cues were viewed as applicable to stimulate the speech sound ($M = 5.29$, $SD =$

1.50), few participants noted that static images limited understanding of the airflow of articulatory movements. They suggested incorporating dynamic or animated cues to illustrate voicing, place, and manner of articulation, particularly for parents facilitating home practice: “When we do speech sound production, we talk about voicing, place, and manner. The ones that are more challenging for families to understand are place and manner” (P6). In contrast, the benefit of other cueing methods, such as minimal pairs and multiple oppositions, was less evident in *MORA*’s current interaction context ($M = 4.0$, $SD = 1.41$). P4 noted, “[system represents] stimulability, yes, but not minimal pair or multiple oppositions.” These findings highlight the importance of developing embodied, multimodal feedback that visualizes articulatory motion and airflow, helping clinicians and caregivers better understand and model speech sound production.

Personalization and educational alignment. Participants valued the ability to configure target words, emphasizing that personalization aligns the tool with therapy goals and children’s everyday contexts: “Use words that children have in their actual life.” (P4) Using words in daily life was perceived to enhance social engagement, reflected in a relatively high score for social skill support ($M = 5.14$, $SD = 1.22$). “It could help with social skills, especially if this is functional for them” (P6). Beyond personalization, they highlighted the benefit of aligning the target with school curricula: “I would want it to match the curriculum. It’s important to me that children can say the words they will encounter in their textbooks.” (P6) Such flexibility supports speech goals and reinforces literacy skills, a view reflected in high ratings for literacy enhancement ($M=5.86$, $SD=1.68$). As P7 summarized, “Having the story, and reading it out loud to them is good for literacy improvement. As it’s read out loud, they’re looking at the words and processing”. Although *MORA* currently focuses on speech production, these findings suggest opportunities to extend its personalization features to more explicitly support social

and literacy development, further strengthening its educational alignment.

5.3.3 RQ3. *How do SLPs perceive and experience the AI-mediated features of MORA?* Overall, participants expressed positive attitudes toward the AI-mediated components of **MORA**. As measured in the TAM survey (Figure 7), they rated the system as ease of learning to use ($M=5.86$, $SD=1.86$), low in mental effort ($M=6.14$, $SD=1.86$), and straightforward to get what they want ($M=5.57$, $SD=1.29$).

Lower the workload. Given their heavy caseloads, SLPs appreciated how **MORA** can reduce the effort for preparation tasks. As P7 noted, *“This is nice because you don’t have to type in any prompt [to LLM] to create a worksheet. It’s automatically generated”*. Additional usage for group therapy in Section 5.3.1 can be one way to reduce efforts by handling multiple children at once. The system’s digital interactivity was also viewed as beneficial for sustaining children’s attention, *“If it involves electronics or the digital world, the more they’re for it, because that just engages them so much better”* (P5). These findings suggest that automation and interactivity can effectively reduce SLPs’ workload while maintaining a sense of control and professional oversight.

Opportunities for social communication and activities. Several participants envisioned extending **MORA**’s conversational interface to include social communication cues or reformatting it to accommodate multiple children joining at once. For example, P3 suggested about the social cues, *“It could be really helpful if, when you’re being polite or rude, it gives you suggestions on how to improve [in situation]”*. Maintaining a narrative and conversational structure, as well as creating turns for multiple children, was another idea to improve social communication significantly. P5 discussed the idea, *“If it’s a group setting, they work together, like everyone is working on the same story and each kid has their own section and record”*. Such ideas highlight potential for AI-mediated dialogue to support social-pragmatic learning, broadening the system’s therapeutic scope beyond articulation practice.

Accessibility and digital literacy. Despite overall enthusiasm, participants expressed concerns about digital literacy among parents and some clinicians who are unfamiliar with digital technology. *“Parents would... You might have to just do a tutorial. For them to really be able to effectively use the app”* (P6), *“Especially clinicians who maybe aren’t as good with technology, they need the work and thought process out of it”* (P5). Conversely, children are considered highly comfortable with digital devices. Previous study [50] shows 96.6% of children use mobile devices before age one and 83% of households own tablets. Designing onboarding flows for less digitally confident adults can enhance accessibility and adoption, leading to parental involvement.

Privacy considerations towards children’s recording. Finally, participants mentioned security concerns surrounding audio data collection of children: P1 pointed out, *“Having their child’s voice recorded in the app, how would you address the confidentiality aspect of that?”*. Given the sensitivity of children’s voice data, the need for strong confidentiality assurances and clear communication on data handling could be the primary design consideration, with reflections on child-centered AI-mediated systems, which must balance data-driven functionality with privacy responsibility.

6 Discussion

6.1 Using AI-mediated Tools to Scale Professional Support

Professional expertise in care and knowledge-intensive domains is intrinsically complex to scale: one expert can only manage a limited number of clients at a time. We suggest that AI-mediated tools enable *scalable expertise* by offloading routine yet critical work (e.g., capture, transcription, triage, and first-pass guidance) while keeping goals and accountability with human professionals. Crucially, such tools *amplify* expertise rather than replace it, like an SLP can effectively reach a big group of children through regular, feedback-driven practice cycles using **MORA**. AI thus expands the surface area where experts can exercise judgment and provide timely, situated support. Similar studies have demonstrated expert systems that address scalability while preserving professional agency in counseling [58] and mental health contexts [51]. Likewise, classroom teachers—who inherently operate in one-to-many environments—can benefit from AI tools that aggregate class data, analyze learning patterns, and reduce administrative workload, enabling reflection and planning for subsequent instruction [42, 67, 89].

Designing for scalable expertise requires more than automating tasks. Systems should keep experts in the loop—providing editability, rationales, and reversible operations—while balancing efficiency with nuance through pre-structured artifacts (e.g., transcripts, summaries, dashboards, and annotations) that let professionals focus on complex or edge cases [51, 58, 102]. These principles extend beyond speech therapy to other one-to-many domains such as mental healthcare and classrooms, where AI-mediated scaffolding can broaden access and sustain care without adding synchronous workload. Ultimately, *scalable expertise* emerges when AI is designed as infrastructure that reallocates expert time and attention rather than substituting for it.

6.2 Designing for Expert Configurability and Trust in AI-mediated Practice Tools

Although SLPs follow similar evidence-based frameworks, each clinician’s practice routine and rationale are highly individualized, aiming to align therapy materials with their existing methods. To accommodate such diversity, **MORA** should provide high configurability—allowing therapists to tailor material parameters (e.g., the amount of auditory bombardment) and data views—while minimizing cognitive effort and preserving interpretability to foster professional trust.

Preset recommendation models can further support efficient configuration by automatically suggesting settings such as repetition levels or auditory bombardment intensity based on a child’s profile, while still allowing experts to fine-tune when needed. However, some participants noted that extensive control might increase cognitive load. For instance, one therapist preferred lightweight automation such as auto-generated weekly summaries, whereas others valued fine-grained filters by syllable, word, or story to examine progress data. Balancing automation and manual control is thus essential to achieve both efficiency and flexibility in expert-oriented design.

This tension between configurability and automation directly connects to professional trust in AI-mediated tools. Trust can be strengthened by enhancing interpretability—clarifying how AI judgments are generated and ensuring they align with clinicians’ goals. For example, simplified indicators such as the number of correctly produced target sounds that can complement the accent and intonation can improve transparency and reduce confusion from probabilistic feedback generated by large language models [66]. As P4 emphasized, *“The app seems very sensitive about the accent. The treatment aims not to focus on the accent, stress, or intonation. Our focus is just to produce sound correctly.”* The system should position AI as a supportive rather than authoritative collaborator in therapeutic decision-making. By providing interpretable feedback and transparent rationales, **MORA** can reinforce professional agency while building sustained trust in AI-assisted clinical practice.

6.3 Scalable Speech Practice to cover Different Disorders with Adaptive Story-based Interactions

While **MORA** is primarily targeted at articulation therapy for SSD, our findings reveal potential to extend its framework to various language and communication disorders. The interactive story-based approach is a flexible scaffold for various therapeutic goals, leveraging the well-established educational value of story-based learning in children’s education [17]. Accommodating customized cues, instructions, and feedback tailored to different disorders can enable us to scale the framework.

SLPs suggested that **MORA**’s approach could cover phonological processes which operate across linguistic contexts, such as final consonant deletion or cluster reduction. The interactive story enables children to practice engagement in connected speech rather than just isolated words, supporting generalization across language levels. Furthermore, comorbidity among speech, language, and voice disorders is reported to be pretty common: 33.9% of those ages 3–10 have multiple communication disorders, while 25.2% of those ages 11–17 have multiple disorders [19], and comorbidity with SSD was 40.8% at 4-year-olds in Australia [36]. Although these disorders have different symptoms, for example, SSD affects speech production accuracy, while DLD impacts semantic and syntactic comprehension, both benefit from practice that promotes generalization. By embedding adaptive dialogue and responsive cues, **MORA**’s story-based design can flexibly scale to these varied needs.

Therefore, this section highlights the broader potential of adaptive story-based interaction as a design pattern for promoting linguistic and social development across disorders.

6.4 Lowering Barriers through Multimodal Interaction

MORA integrates multimodal interaction—visual, auditory, and embodied cues—to support pre-literate children’s expressive engagement. Here, we discuss how such interaction can further enhance effectiveness and engagement.

Speech sound therapy relies on physical cues, such as mouth models or short video demonstrations, which can enhance comprehension and limitations [28]. Consistent with prior work on multimodal instruction [70], our formative study also found that

physical demonstrations of mouth movements and hand gesture cues improved children’s accuracy and motivation. Integrating mouth-shape animations at the word or syllable level could also depict complex coarticulations (assimilation effects), providing a more intuitive way to understand continuous sound production. Because young children are often pre-literate, spoken input and auditory feedback can replace text-based instructions [61]. Simple, literal icons paired with sound afford intuitive navigation and reduce cognitive load [87, 101]. Such design considerations are essential for maintaining usability and engagement in early childhood learning environments.

These multimodal design principles make speech practice more accessible and engaging, bridging therapeutic rigor with play-based learning. To be truly effective, digital tools must guide children through graded levels of speech practice while sustaining their attention and motivation. Embedding practice targets into character-driven narratives offers a promising direction, allowing for repeated use of key sounds and words in increasingly complex linguistic units, all while maintaining engagement through plot progression and role-playing.

6.5 Limitations and Future Work

While our findings highlight **MORA**’s potential to enhance speech practice and clinician–child collaboration, several limitations inform directions for future research. First, the current study evaluated **MORA** primarily through expert feedback from licensed SLPs rather than direct use by children or families. Although this expert-centered approach was valuable for validating therapeutic alignment and design feasibility, future studies should include longitudinal deployments with children and caregivers to assess learning outcomes, sustained engagement, and real-world usability. Such in-situ evaluations would provide richer evidence of **MORA**’s effectiveness in supporting home practice and generalization of speech sounds beyond controlled settings.

Second, the present prototype focuses on a limited set of target phonemes and narrative templates designed in collaboration with SLPs. Expanding these resources to cover a broader range of phonological processes, difficulty levels, and cultural contexts will improve inclusivity and scalability. Future iterations should also explore adaptive personalization based on each child’s performance history and motivational patterns—such as dynamically adjusting story complexity, feedback frequency, or reward mechanisms. Finally, while **MORA** currently supports asynchronous therapist review, integrating real-time or semi-automated feedback loops (e.g., AI-assisted scoring verified by SLPs) could further reduce clinician workload and strengthen the AI-in-the-loop workflow.

These directions highlight opportunities to extend **MORA** from a research prototype to a clinically deployable platform that continuously learns from real-world use, bridges professional and home environments, and contributes to more accessible and engaging digital speech therapy for early childhood development.

7 Conclusion

We presented **MORA**, an interactive story-based conversational practice system designed to support SSD home-based practice of SSD children. Through iterative design and an expert study with

seven experienced and licensed SLPs, we demonstrated that **MORA** aligns closely with established clinical practices while introducing new opportunities for engaging, AI-mediated home practice. Our findings underscore the importance of designing AI-mediated therapeutic systems that strike a balance between expert configurability and automation, transparency, and trust. **MORA**'s combination of story-based interaction, phoneme-level analysis, and multimodal feedback exemplifies how technology can extend evidence-based speech therapy into naturalistic, accessible contexts for children and their parents. Looking forward, this work opens pathways for longitudinal and telehealth deployments, examining how adaptive personalization and multimodal interaction can further enhance learning and generalization across diverse communication disorders. Beyond SSD, these research contributions broaden understanding of how AI-in-the-loop systems can augment professional expertise while supporting inclusive, child-centered therapeutic experiences.

8 GenAI Usage Disclosure

We used programming support from Claude-4-sonnet and Claude-4.5-sonnet in Cursor in the implementation of the system. Additionally, we used writing support from Grammarly and ChatGPT to correct grammatical errors and polish the writing of the paragraphs.

References

- [1] 2022. Childhood Apraxia of Speech Therapy | Apraxia Farm App 2022. <https://www.smartyearsapps.com/apraxia-farm/>
- [2] 2025. ATLAS.ti | The #1 Software for Qualitative Data Analysis - ATLAS.ti. <https://atlasti.com/>
- [3] 2025. The CMU Pronouncing Dictionary. <http://www.speech.cs.cmu.edu/cgi-bin/cmudict#about>
- [4] 2025. Employment Settings for SLPs. <https://www.asha.org/students/employment-settings-for-slp/> Publisher: American Speech-Language-Hearing Association.
- [5] 2025. ginic/gender_split_70_female_1_wav2vec2-large-xlsr-53-buckeye-ipa · Hugging Face. https://huggingface.co/ginic/gender_split_70_female_1_wav2vec2-large-xlsr-53-buckeye-ipa
- [6] 2025. KoelLabs/xlsr-timit-b0 · Hugging Face. <https://huggingface.co/KoelLabs/xlsr-timit-b0>
- [7] 2025. Little Stories Pro - App for speech, language and literacy for iPad by Little Bee Speech. https://littlebeespeech.com/little_stories_pro.php
- [8] 2025. R-colored vowel. https://en.wikipedia.org/w/index.php?title=R-colored_vowel&oldid=1310609349 Page Version ID: 1310609349.
- [9] 2025. /r/-Controlled Vowels. <https://tfcs.baruch.cuny.edu/r-controlled-vowels/>
- [10] 2025. Speech Sound Disorders: Articulation and Phonology. <https://www.asha.org/practice-portal/clinical-topics/articulation-and-phonology/> Publisher: American Speech-Language-Hearing Association.
- [11] 2025. Spokelt. <https://spokeitthegame.com/>
- [12] 2025. STAR: Multimedia speech databases. <https://www.seeingsspeech.ac.uk/speechstar/disordered-speech-sentences-database/>
- [13] Melissa M. Allen. 2013. Intervention Efficacy and Intensity for Children With Speech Sound Disorder. *Journal of Speech, Language, and Hearing Research* 56, 3 (June 2013), 865–877. [https://doi.org/10.1044/1092-4388\(2012/11-0076\)](https://doi.org/10.1044/1092-4388(2012/11-0076)) Publisher: American Speech-Language-Hearing Association.
- [14] Rafiah Ansari, Ros Herman, and Martin Cartwright. 2025. Designing a digital intervention for children with developmental language disorder: Aligning theory, research and clinical practice. *DIGITAL HEALTH* 11 (May 2025), 20552076251324442. <https://doi.org/10.1177/20552076251324442> Publisher: SAGE Publications Ltd.
- [15] Rafiah Ansari, Elena Lorraine, and Mélanie Gréaux. 2022. Impact of COVID-19 on digital practice in UK paediatric speech and language therapy and implications for the future: A national survey. *International Journal of Language & Communication Disorders* 57, 5 (2022), 1112–1129. <https://doi.org/10.1111/1460-6984.12750> _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/1460-6984.12750>
- [16] Rachel G Barnett and Paul W Casella. 2024. A Qualitative Evaluation of Treatment App Features for Children With Speech Sound Disorders. *Communication Disorders Quarterly* (2024), 15257401241299719.
- [17] Murat Barta. 2020. The Use of Storytelling Methods by Teachers and Their Effects on Children's Understanding and Attention Span. *Southeast Asia Early Childhood* 9, 1 (2020), 75–84. <https://eric.ed.gov/?id=EJ1272217> Publisher: National Child Development Research Centre ERIC Number: EJ1272217.
- [18] John E. Bernthal, Nicholas W. Bankson, and Peter Flipsen. 2009. *Articulation and Phonological Disorders: Speech Sound Disorders in Children*. Pearson/Allyn & Bacon. Google-Books-ID: j_UmKAAACAAJ.
- [19] Lindsey I. Black, Anjel Vahratian, and Howard J. Hoffman. 2015. Communication Disorders and Use of Intervention Services Among Children Aged 3-17 Years: United States, 2012. *NCHS data brief* 205 (June 2015), 1–8.
- [20] J. Martin Bland and Douglas G. Altman. 1997. Statistics notes: Cronbach's alpha. *BMJ* 314, 7080 (Feb. 1997), 572. <https://doi.org/10.1136/bmj.314.7080.572> Publisher: British Medical Journal Publishing Group Section: General practice.
- [21] Gordon W. Blood, Jenna Swavely Ridenour, Emily A. Thomas, Constance Dean Qualls, and Carol Scheffner Hammer. 2002. Predicting Job Satisfaction Among Speech-Language Pathologists Working in Public Schools. *Language, Speech, and Hearing Services in Schools* 33, 4 (Oct. 2002), 282–290. [https://doi.org/10.1044/0161-1461\(2002/023\)](https://doi.org/10.1044/0161-1461(2002/023))
- [22] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (Jan. 2006), 77–101. <https://doi.org/10.1191/1478088706qp0630a> Publisher: Routledge _eprint: <https://doi.org/10.1191/1478088706qp0630a>
- [23] Virginia Braun and Victoria Clarke. 2019. Reflecting on reflexive thematic analysis. *Qualitative Research in Sport, Exercise and Health* 11, 4 (Aug. 2019), 589–597. <https://doi.org/10.1080/2159676X.2019.1628806> Publisher: Routledge.
- [24] Jan Broomfield and Barbara Dodd. 2004. Children with speech and language disability: caseload characteristics. *International Journal of Language & Communication Disorders* 39, 3 (2004), 303–324. <https://doi.org/10.1080/13682820310001625589> _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1080/13682820310001625589>
- [25] Kathryn Cabbage, Kelly Farquharson, and Shari DeVeney. 2022. Speech sound disorder treatment approaches used by school-based clinicians: An application of the experience sampling method. *Language, Speech, and Hearing Services in Schools* 53, 3 (2022), 860–873.
- [26] Kathryn L. Cabbage and Elaine R. Hitchcock. 2022. Clinical Considerations for Speech Perception in School-Age Children With Speech Sound Disorders: A Review of the Current Literature. *Language, Speech, and Hearing Services in Schools* 53, 3 (July 2022), 768–785. https://doi.org/10.1044/2022_LSHSS-21-00120 Publisher: American Speech-Language-Hearing Association.
- [27] Philip N Combiths, S. Pruitt-Lord, A. Escobedo, and J. Barlow. 2021. Phonological complexity in intervention for Spanish-speaking children with speech sound disorder. *Clinical Linguistics Phonetics* 36 (2021), 219 – 240. <https://doi.org/10.1080/02699206.2021.1936186>
- [28] Aayushi Dangol, Aaleyah Lewis, Hyewon Suh, Xuesi Hong, Hedda Meadan, James Fogarty, and Julie A. Kientz. 2025. "I Want to Think Like an SLP": A Design Exploration of AI-Supported Home Practice in Speech Therapy. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 1–22. <https://doi.org/10.1145/3706598.3713986>
- [29] Ali Darejeh, Siti Salwah Salim, and Adeleh Asemi. 2019. Speech pronunciation practice system for speech-impaired children: a systematic review of impacts and functionality. *Universal Access in the Information Society* 18, 1 (March 2019), 169–189. <https://doi.org/10.1007/s10209-017-0573-5>
- [30] Nathan Denny. 2024. Behind the SLP Shortage: Meeting Speech Pathologist Demand. <https://www.careerstaff.com/healthcare-staffing-blog/healthcare-staffing/slp-shortage-meeting-speech-pathologist-demand/>
- [31] Wendy Doubé, Paul Carding, Kieran Flanagan, Jordy Kaufman, and Hannah Armitage. 2018. Comparing feedback types in multimedia learning of speech by young children with common speech sound disorders: Research protocol for a pretest posttest independent measures control trial. *Frontiers in Psychology* 9 (2018), 444.
- [32] Yao Du and Felix Juefei-Xu. 2023. Generative AI for Therapy? Opportunities and Barriers for ChatGPT in Speech-Language Therapy. (March 2023). <https://openreview.net/forum?id=CRZSr6TprIS>
- [33] Yao Du, Kathryn Lubniewski, Lori Price, Grace Breslin, Paula Thomson, Natasha Jinadasa, and Nikita Soni. 2023. "They Can't Believe They're a Tiger": Insights from pediatric speech-language pathologist mobile app users and app designers. *International Journal of Language & Communication Disorders* 58, 5 (2023), 1717–1737.
- [34] Yao Du and Katie Salen Tekinbas. 2020. Bridging the gap in mobile interaction design for children with disabilities: Perspectives from a pediatric speech language pathologist. *International Journal of Child-Computer Interaction* 23-24 (June 2020), 100152. <https://doi.org/10.1016/j.ijcci.2019.100152>
- [35] Jared Duval. 2017. A mobile game system for improving the speech therapy experience. In *Proceedings of the 19th International Conference on Human-Computer Interaction with Mobile Devices and Services*. 1–3.
- [36] Patricia Eadie, Angela Morgan, Obioha C Ukoumunne, Kyriaki Ttofari Eecen, Melissa Wake, and Sheena Reilly. 2015. Speech sound disorder at 4

- years: prevalence, comorbidities, and predictors in a community cohort of children. *Developmental Medicine & Child Neurology* 57, 6 (2015), 578–584. <https://doi.org/10.1111/dmcn.12635> eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/dmcn.12635>.
- [37] JaNeal Freeman. 2016. Speaking of Stories: An Exploration of Oral Stories as an Intervention in Speech Sound Disorders on JSTOR. *Storytelling, Self, Society* 12, 2 (2016), 207–247. <https://www.jstor.org/stable/10.13110/storselfsoci.12.2.0207?seq=17>
- [38] JaNeal B Freeman. 2016. Speaking of Stories: An Exploration of Oral Stories as an Intervention in Speech Sound Disorders. *Storytelling, Self, Society* 12, 2 (2016), 207–247.
- [39] Lisa Furlong, Meg Morris, Tanya Serry, and Shane Erickson. 2018. Mobile apps for treatment of speech disorders in children: An evidence-based analysis of quality and efficacy. *PloS one* 13, 8 (2018), e0201513.
- [40] Mateja Gačnik, Andreja Istenič Starčič, Janez Zaletelj, and Matej Zajc. 2018. User-centred app design for speech sound disorders interventions with tablet computers. *Universal Access in the Information Society* 17, 4 (2018), 821–832.
- [41] Amy M Glaspey, Jenica J Wilson, Justin D Reeder, W. Tseng, and A. Macleod. 2022. Moving Beyond Single Word Acquisition of Speech Sounds to Connected Speech Development With Dynamic Assessment. *Journal of speech, language, and hearing research : JSLHR* (2022), 1–17. https://doi.org/10.1044/2021_JSLHR-21-00188
- [42] Elena L. Glassman, Jeremy Scott, Rishabh Singh, Philip J. Guo, and Robert C. Miller. 2015. OverCode: Visualizing Variation in Student Solutions to Programming Problems at Scale. *ACM Trans. Comput.-Hum. Interact.* 22, 2 (2015), 7:1–7:35. <https://doi.org/10.1145/2699751>
- [43] Rebecca Goodhue, Mark Onslow, Susan Quine, Sue O'Brian, and Anna Hearne. 2010. The Lidcombe Program of early stuttering intervention: mothers' experiences. *Journal of Fluency Disorders* 35, 1 (March 2010), 70–84. <https://doi.org/10.1016/j.jfludis.2010.02.002>
- [44] Leo A. Goodman. 1961. Snowball Sampling. *The Annals of Mathematical Statistics* 32, 1 (1961), 148–170. <http://www.jstor.org/stable/2237615> Publisher: Institute of Mathematical Statistics.
- [45] Jessica Gosnell. 2011. Apps: An Emerging Tool for SLPs: A plethora of apps can be used to develop expressive, receptive, and other language skills. *The ASHA Leader* 16, 12 (2011), 10–13.
- [46] Adam Hair, Kirrie J. Ballard, Constantina Markoulli, Penelope Monroe, Jacqueline McKechnie, Beena Ahmed, and Ricardo Gutierrez-Osuna. 2021. A Longitudinal Evaluation of Tablet-Based Child Speech Therapy with Apraxia World. *ACM Trans. Access. Comput.* 14, 1 (March 2021), 3:1–3:26. <https://doi.org/10.1145/3433607>
- [47] Adam Hair, Penelope Monroe, Beena Ahmed, Kirrie J Ballard, and Ricardo Gutierrez-Osuna. 2018. Apraxia world: A speech therapy game for children with speech sound disorders. In *Proceedings of the 17th ACM Conference on Interaction Design and Children*. 119–131.
- [48] Nikki Heyman. 2020. Identifying features of apps to support using evidence-based language intervention with children. *Assistive Technology* (2020).
- [49] Luis M. T. Jesus, Joana Martinez, Joaquim Santos, Andreia Hall, and V. Joffe. 2019. Comparing Traditional and Tablet-Based Intervention for Children With Speech Sound Disorders: A Randomized Controlled Trial. *Journal of speech, language, and hearing research : JSLHR* (2019), 1–17. https://doi.org/10.1044/2019_jslhr-s-18-0301
- [50] Hilda K. Kabali, Matilde M. Irigoyen, Rosemary Nunez-Davis, Jennifer G. Budacki, Sweta H. Mohanty, Kristin P. Leister, and Robert L. Bonner, Jr. 2015. Exposure and Use of Mobile Media Devices by Young Children. *Pediatrics* 136, 6 (Dec. 2015), 1044–1050. <https://doi.org/10.1542/peds.2015-2151>
- [51] Jiwon Kim, Jiwon Kang, Taeun Kim, Hayeon Song, and Jinyoung Han. 2023. AlphaDAPR: An AI-based Explainable Expert Support System for Art Therapy. In *Proceedings of the 28th International Conference on Intelligent User Interfaces (IUI '23)*. Association for Computing Machinery, New York, NY, USA, 19–31. <https://doi.org/10.1145/3581641.3584087>
- [52] Eunhye Grace Ko, Rotem Landesman, Jason C Young, Ahmer Arif, Katie Davis, and Angela D. R. Smith. 2025. Domain Experts, Design Novices: How Community Practitioners Enact Participatory Design Values. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3706598.3714060>
- [53] Breanna I. Krueger. 2019. Eligibility and Speech Sound Disorders: Assessment of Social Impact. *Perspectives of the ASHA Special Interest Groups* 4, 1 (Feb. 2019), 85–90. https://doi.org/10.1044/2018_PERS-SIG1-2018-0016 Publisher: American Speech-Language-Hearing Association.
- [54] C. H. Lawshe. 1975. A QUANTITATIVE APPROACH TO CONTENT VALIDITY¹. *Personnel Psychology* 28, 4 (Dec. 1975), 563–575. <https://doi.org/10.1111/j.1744-6570.1975.tb01393.x>
- [55] Naomi Leafe, Emma Pagnamenta, Mark Donnelly, Laurence Taggart, and J. Titterton. 2025. "To Know That You Are a Link in the Chain": A Realist Evaluation to Explore How Digital, Intensive, Parent-Implemented Interventions Work for Children With Speech Sound Disorder, Why, and for Whom. *International Journal of Language Communication Disorders* 60 (2025). <https://doi.org/10.1111/1460-6984.70049>
- [56] Naomi Leafe, Emma Pagnamenta, Laurence Taggart, M. Donnelly, A. Hassiotis, and J. Titterton. 2024. What works, how and in which contexts when supporting parents to implement intensive speech and language therapy at home for children with speech sound disorder? A protocol for a realist review. *BMJ Open* 14 (2024). <https://doi.org/10.1136/bmjopen-2023-074272>
- [57] Naomi Leafe, Emma Pagnamenta, Laurence Taggart, Mark Donnelly, Angela Hassiotis, and J. Titterton. 2025. What works, how and in which contexts when using digital health to support parents/carers to implement intensive speech and language therapy at home for children with speech sound disorder? A realist review. *PLOS One* 20 (2025). <https://doi.org/10.1371/journal.pone.0321647>
- [58] Hyunjung Lee, Jiyeon Lee, Migyeong Yang, Daeun Lee, Hayeon Song, Youjin Han, and Jinyoung Han. 2025. Counselor-AI Collaborative Transcription and Editing System for Child Counseling Analysis. In *Proceedings of the 30th International Conference on Intelligent User Interfaces (IUI '25)*. Association for Computing Machinery, New York, NY, USA, 425–445. <https://doi.org/10.1145/3708359.3712081>
- [59] Pascal Lefebvre, Alexandra Fiorino, Dana Johannsen, Madeleine Tait, Andrea Tkalec, and Ann Sutton. 2017. Shared Storybook Reading to Enhance Early Literacy Skills of Children with Speech Sound Disorders: A Feasibility Study. *Canadian Journal of Speech-Language Pathology & Audiology* 41 (Jan. 2017), 72–91. <https://research.ebsco.com/linkprocessor/plink?id=74837fdb-2d73-3836-961a-ae46b50b85d3> Publisher: Speech-Language & Audiology Canada.
- [60] Dancheng Liu, Jason Yang, Ishan Albrecht-Buehler, Helen Qin, Sophie Li, Yuting Hu, Amir Nassereldine, and Jinjun Xiong. 2024. Automatic Screening for Children with Speech Disorder Using Automatic Speech Recognition: Opportunities and Challenges. *Proceedings of the AAAI Symposium Series* 4, 1 (Nov. 2024), 308–313. <https://doi.org/10.1609/aaais.v4i1.31807>
- [61] Silvia B. Lovato and Anne Marie Piper. 2019. Young Children and Voice Search: What We Know From Human-Computer Interaction Research. *Frontiers in Psychology* 10 (Jan. 2019). <https://doi.org/10.3389/fpsyg.2019.00008> Publisher: Frontiers.
- [62] Tanya Matthews, Alexandra Barbeau-Morrison, and S. Rvachew. 2021. Application of the Challenge Point Framework During Treatment of Speech Sound Disorders. *Journal of speech, language, and hearing research : JSLHR* (2021), 1–17. https://doi.org/10.1044/2021_JSLHR-20-00437
- [63] Sharynne McLeod, Graham Daniel, and Jacqueline Barr. 2013. "When he's around his brothers ... he's not so quiet": The private and public worlds of school-aged children with speech sound disorder. *Journal of Communication Disorders* 46, 1 (Jan. 2013), 70–83. <https://doi.org/10.1016/j.jcomdis.2012.08.006>
- [64] Sharynne McLeod and Linda J. Harrison. 2009. Epidemiology of Speech and Language Impairment in a Nationally Representative Sample of 4- to 5-Year-Old Children. *Journal of Speech, Language, and Hearing Research* 52, 5 (Oct. 2009), 1213–1229. [https://doi.org/10.1044/1092-4388\(2009/08-0085\)](https://doi.org/10.1044/1092-4388(2009/08-0085)) Publisher: American Speech-Language-Hearing Association.
- [65] David R. Mortensen, Patrick Littell, Akash Bharadwaj, Kartik Goyal, Chris Dyer, and Lori Levin. 2016. PanPhon: A Resource for Mapping IPA Segments to Articulatory Feature Vectors. In *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, Yuji Matsumoto and Rashmi Prasad (Eds.). The COLING 2016 Organizing Committee, Osaka, Japan, 3475–3484. <https://aclanthology.org/C16-1328/>
- [66] Aliakbar Nafar, Kristen Brent Venable, and Parisa Kordjamshidi. 2024. Reasoning over Uncertain Text by Generative Large Language Models. <https://doi.org/10.48550/arXiv.2402.09614> arXiv:2402.09614 [cs].
- [67] Tricia J. Ngoon, S Sushil, Angela E.B. Stewart, Ung-Sang Lee, Saranya Venkatraman, Neil Hawani, Prasenjit Mitra, Sherice Clarke, John Zimmerman, and Amy Ogan. 2024. ClassInSight: Designing Conversation Support Tools to Visualize Classroom Discussion for Personalized Teacher Professional Development. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3613904.3642487>
- [68] Lawrence A. Palinkas, Sarah M. Horwitz, Carla A. Green, Jennifer P. Wisdom, Naihua Duan, and Kimberly Hoagwood. 2015. Purposeful Sampling for Qualitative Data Collection and Analysis in Mixed Method Implementation Research. *Administration and Policy in Mental Health and Mental Health Services Research* 42, 5 (Sept. 2015), 533–544. <https://doi.org/10.1007/s10488-013-0528-y>
- [69] Rhea Paul and Paul Cascella. 2014. Ethical Practice in Communication Disorders. *Communication Disorders Faculty Publications* (Jan. 2014). https://digitalcommons.sacredheart.edu/speech_fac/149
- [70] Robert A. Pieretti, Sandra D. Kaul, Razi M. Zarchy, and Lauren M. O'Hanlon. 2015. Using a Multimodal Approach to Facilitate Articulation, Phonemic Awareness, and Literacy in Young Children. *Communication Disorders Quarterly* 36,

- 3 (May 2015), 131–141. <https://doi.org/10.1177/1525740114545360> Publisher: SAGE Publications Inc.
- [71] Denise F. Polit and Cheryl Tatano Beck. 2006. The content validity index: Are you sure you know what's being reported? critique and recommendations. *Research in Nursing & Health* 29, 5 (Oct. 2006), 489–497. <https://doi.org/10.1002/nur.20147>
- [72] Denise F. Polit, Cheryl Tatano Beck, and Steven V. Owen. 2007. Is the CVI an acceptable indicator of content validity? Appraisal and recommendations. *Research in Nursing & Health* 30, 4 (2007), 459–467. <https://doi.org/10.1002/nur.20199> _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/nur.20199>
- [73] Lucy Rodgers, Nicola Botting, Martin Cartwright, Sam Harding, and Rosalind Herman. 2023. Shared characteristics of intervention techniques for oral vocabulary and speech comprehensibility in preschool children with co-occurring features of developmental language disorder and a phonological speech sound disorder: protocol for a systematic review with narrative synthesis. *BMJ open* 13, 6 (2023), e071262.
- [74] William R. Rodríguez, Oscar Saz, and Eduardo Lleida. 2012. A prelingual tool for the education of altered voices. *Speech Commun.* 54, 5 (June 2012), 583–600. <https://doi.org/10.1016/j.specom.2011.05.006>
- [75] Heather Leavy Rusiewicz and Jessica Lynch Rivera. 2017. The Effect of Hand Gesture Cues Within the Treatment of /r/ for a College-Aged Adult With Persisting Childhood Apraxia of Speech. *American Journal of Speech-Language Pathology* 26, 4 (Nov. 2017), 1236–1243. https://doi.org/10.1044/2017_AJSLP-15-0172 Publisher: American Speech-Language-Hearing Association.
- [76] Joaquim Santos, Mário Vairinhos, and Luis MT Jesus. 2019. Treating children with speech sound disorders: development of a tangible artefact prototype. *JMIR Serious Games* 7, 4 (2019), e13861.
- [77] Jo Saul and Courtenay Norbury. 2020. Feasibility of an app-based parent-mediated speech production intervention for minimally verbal autistic children: development and pilot testing of a new intervention. *Pilot and feasibility studies* 6, 1 (2020), 185.
- [78] William Seymour, Mark Cote, and Jose Such. 2023. Legal Obligation and Ethical Best Practice: Towards Meaningful Verbal Consent for Voice Assistants. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3544548.3580967>
- [79] Zhonghao Shi, Daeun Chung, Yao Du, Jiaxin Zhang, Shivani Raina, and Maja Mataric. 2025. Is AI Ready to Support Speech Therapy for Children? A Systematic Review of AI-Enabled Mobile Apps for Pediatric Speech Therapy. *Proceedings of the 24th Interaction Design and Children (2025)*, 479–493.
- [80] Lawrence D. Shriberg, J. Bruce Tomblin, and Jane L. McSweeney. 1999. Prevalence of Speech Delay in 6-Year-Old Children and Comorbidity With Language Impairment. *Journal of Speech, Language, and Hearing Research* 42, 6 (Dec. 1999), 1461–1481. <https://doi.org/10.1044/jslhr.4206.1461> Publisher: American Speech-Language-Hearing Association.
- [81] H. Storkel. 2022. Minimal, Maximal, or Multiple: Which Contrastive Intervention Approach to Use With Children With Speech Sound Disorders? *Language, speech, and hearing services in schools* (2022), 1–14. https://doi.org/10.1044/2021_LSHSS-21-00105
- [82] Eleanor Sugden, Elise Baker, Natalie Munro, and A. Lynn Williams. 2016. Involvement of parents in intervention for childhood speech sound disorders: a review of the evidence. *International Journal of Language & Communication Disorders* 51, 6 (2016), 597–625. <https://doi.org/10.1111/1460-6984.12247> _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/1460-6984.12247>
- [83] Eleanor Sugden, Elise Baker, Natalie Munro, A. Lynn Williams, and Carol M. Trivette. 2018. An Australian survey of parent involvement in intervention for childhood speech sound disorders. *International Journal of Speech-Language Pathology* 20, 7 (Dec. 2018), 766–778. <https://doi.org/10.1080/17549507.2017.1356936> Publisher: Taylor & Francis _eprint: <https://doi.org/10.1080/17549507.2017.1356936>
- [84] Eleanor Sugden, Elise Baker, A. Lynn Williams, Natalie Munro, and Carol M. Trivette. 2020. Evaluation of Parent- and Speech-Language Pathologist-Delivered Multiple Oppositions Intervention for Children With Phonological Impairment: A Multiple-Baseline Design Study. *American Journal of Speech-Language Pathology* 29, 1 (Feb. 2020), 111–126. https://doi.org/10.1044/2019_AJSLP-18-0248 Publisher: American Speech-Language-Hearing Association.
- [85] Eleanor Sugden, Natalie Munro, Carol M. Trivette, Elise Baker, and A. Lynn Williams. 2019. Parents' Experiences of Completing Home Practice for Speech Sound Disorders. *Journal of Early Intervention* 41, 2 (June 2019), 159–181. <https://doi.org/10.1177/1053815119828409> Publisher: SAGE Publications Inc.
- [86] Hyewon Suh, Aayushi Dangol, Hedda Meadan, Carol A. Miller, and Julie A. Kientz. 2024. Opportunities and Challenges for AI-Based Support for Speech-Language Pathologists. In *Proceedings of the 3rd Annual Meeting of the Symposium on Human-Computer Interaction for Work (CHIWORK '24)*. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3663384.3663387>
- [87] Sulakshana. 2022. Design considerations for kids. <https://medium.com/design-bootcamp/design-considerations-for-kids-48ec9bf2b18>
- [88] Sherine R. Tambyraja, Kelly Farquharson, and Laura Justice. 2020. Reading Risk in Children With Speech Sound Disorder: Prevalence, Persistence, and Predictors. *Journal of Speech, Language, and Hearing Research* 63, 11 (Nov. 2020), 3714–3726. https://doi.org/10.1044/2020_JSLHR-20-00108 Publisher: American Speech-Language-Hearing Association.
- [89] Xiaohang Tang, Sam Wong, Marcus Huynh, Zicheng He, Yalong Yang, and Yan Chen. 2025. SPHERE: Supporting Personalized Feedback at Scale in Programming Classrooms with Structured Review of Generative AI Outputs. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Association for Computing Machinery, New York, NY, USA, 1–17. <https://doi.org/10.1145/3706599.3720203>
- [90] Veronica A. Thurmond. 2001. The Point of Triangulation. *Journal of Nursing Scholarship* 33, 3 (2001), 253–258. <https://doi.org/10.1111/j.1547-5069.2001.00253.x> _eprint: <https://sigmapubs.onlinelibrary.wiley.com/doi/pdf/10.1111/j.1547-5069.2001.00253.x>
- [91] Jimmy Tobin, Phillip Nelson, Bob MacDonald, Rus Heywood, Richard Cave, Katie Seaver, Antoine Desjardins, Pan-Pan Jiang, and Jordan R. Green. 2024. Automatic Speech Recognition of Conversational Speech in Individuals With Disordered Speech. *Journal of Speech, Language, and Hearing Research* 67, 11 (Nov. 2024), 4176–4185. https://doi.org/10.1044/2024_JSLHR-24-00045 Publisher: American Speech-Language-Hearing Association.
- [92] Rachel Tosh, Wendy Arnott, and Nerina Scarinci. 2017. Parent-implemented home therapy programmes for speech and language: a systematic review. *International Journal of Language & Communication Disorders* 52, 3 (2017), 253–269. <https://doi.org/10.1111/1460-6984.12280> _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/1460-6984.12280>
- [93] Atiyeh Vaezipour, Jessica Campbell, Deborah Theodoros, Trevor Russell, et al. 2020. Mobile apps for speech-language therapy in adults with communication disorders: review of content and quality. *JMIR mHealth and uHealth* 8, 10 (2020), e18858.
- [94] Viswanath Venkatesh and Hillol Bala. 2008. Technology Acceptance Model 3 and a Research Agenda on Interventions. *Decision Sciences* 39, 2 (2008), 273–315. <https://doi.org/10.1111/j.1540-5915.2008.00192.x> _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1540-5915.2008.00192.x>
- [95] Nicole Watts Pappas, Lindy McAllister, and Sharynne McLeod. 2016. Parental beliefs and experiences regarding involvement in intervention for their child with speech sound disorder. *Child Language Teaching and Therapy* 32, 2 (June 2016), 223–239. <https://doi.org/10.1177/0265659015615925> Publisher: SAGE Publications Ltd.
- [96] Rachel L Wellman, Barbara A Lewis, Lisa A Freebairn, Allison A Avrich, Amy J Hansen, and Catherine M Stein. 2011. Narrative ability of children with speech sound disorders and the prediction of later literacy skills. *Narrative* (2011).
- [97] A. Lynn Williams, Sharynne McLeod, and Rebecca J. McCauley. 2010. *Interventions for Speech Sound Disorders in Children*. Brookes Publishing Company. Publication Title: Brookes Publishing Company ERIC Number: ED509560.
- [98] Yvonne Wren, Sam Harding, Juliet Goldbart, and Sue Roulstone. 2018. A systematic review and classification of interventions for speech-sound disorder in preschool children. *International Journal of Language & Communication Disorders* 53, 3 (2018), 446–467. <https://doi.org/10.1111/1460-6984.12371> _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/1460-6984.12371>
- [99] Yvonne Wren, Laura L. Miller, Tim J. Peters, Alan Emond, and Sue Roulstone. 2016. Prevalence and Predictors of Persistent Speech Sound Disorder at Eight Years Old: Findings From a Population Cohort Study. *Journal of Speech, Language, and Hearing Research* 59, 4 (Aug. 2016), 647–673. https://doi.org/10.1044/2015_JSLHR-S-14-0282 Publisher: American Speech-Language-Hearing Association.
- [100] Yvonne Wren, Emma Pagnamenta, Faith Orchard, Tim J. Peters, Alan Emond, Kate Northstone, Laura Louise Miller, and Susan Roulstone. 2023. Social, emotional and behavioural difficulties associated with persistent speech disorder in children: A prospective population study. *JCPP Advances* 3, 1 (2023), e12126. <https://doi.org/10.1002/jcv2.12126> _eprint: <https://acamh.onlinelibrary.wiley.com/doi/pdf/10.1002/jcv2.12126>
- [101] Ying Xu, Stacy Branham, Xinwei Deng, Penelope Collins, and Mark Warschauer. 2021. Are Current Voice Interfaces Designed to Support Children's Language Development?. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3411764.3445271>
- [102] Qingxiao Zheng, Parisa Rabbani, Yu-Rou Lin, Daan Mansour, and Yun Huang. 2024. SOAP: AI: A Collaborative Tool for Documenting Human Behavior in Videos through Multimodal Generative AI. In *Companion Publication of the 2024 Conference on Computer-Supported Cooperative Work and Social Computing*. 87–90.